

Linux Run Levels

The run level is a specific single-digit integer that helps define the operating state of the Linux-based operating system. However, each run level is delegated to various system configurations and allows to access various process combinations.

In this reading, you'll be able to describe the run levels in Linux.

What are run levels?

The Run levels are the state of the init command and define the system services for which they operate. Numbers are useful for identifying run levels. Run levels play an important role in managing the Linux system.

Run level in Linux is a mode of operation that defines the services and processes that are available for users and the systems. Each run level defines specific services and is designed to perform different tasks.

Let's look at the table below to understand various Linux run levels.

Run levels	Level name	Description
0	Halt the system	Stop all system operations
1	Single-user mode	Allows only one user to maintain the tasks
2	Multi-user mode without network file system (NFS)	Multiple users can access the system; however, they cannot share files over the network
3	Full multi-user mode (text-based)	Multiple users with full capabilities, however, cannot access graphics
4	Undefined or not used (reserved for user-defined settings)	No default function is available; however, users can customize
5	Full multi-user mode (with graphical user interface)	Multiple users can access with a graphical interface for easier use
6	Reboot the system	Restarts the system; you can refresh all operations

Example of Run Levels

Consider an administrator who wants to perform system maintenance that requires system access for the root user. For this, the administrator would use run level 1, single-user mode. Let's look at the step-wise process the administrator would follow to use Linux at the run level. This process ensures that maintenance and updates occur without disrupting user access or risking system integrity.

1. Log in as the root user or a user with the sudo privileges role because changing run levels requires administrative permissions.
2. Next, access the terminal window on the Linux system if the system is running in the graphical interface.
3. Use the init command to change the run level. If the user has not logged in as a root user, the administrator would enter **init 1** or **sudo init 1** in the terminal to move to single-user mode. The init command prompts the system to terminate all processes and services that are not required for the single-user mode. Further, it will switch the system to run level 1.
4. Now, perform the necessary system maintenance tasks. In single-user mode, the administrator has allowed access to all file systems and commands without networking services.
5. Finally, return the system to its normal operational state after completing the maintenance tasks. The administrator would use the command init 5 to return to full multi-user mode with a graphical interface when the system operates in the graphical mode. However, the administrator would use init 3 when the systems are usually in a text-based multi-user mode.

How do run levels work?

Once the system completes the Linux system boot process, the kernel initiates the hardware and mounts the filesystems. Further, it passes control to the 'init' process, which is the first process from where Linux starts its process.

At this point, the system transitions from its initial boot state to the operational state by moving into one of the several predetermined run levels. These run levels define the types of services or processes that would be achieved to meet the system's or its users' requirements.

Summary

In this reading, you've learned about the Linux run levels. Run level is a preset single-digit integer used to define the operating state of your Linux-based operating system.

The Linux run levels have seven levels such as:

- Run level 0: Halt the system
- Run level 1: Single-user mode
- Run level 2: Multi-user mode without network file system (NFS)
- Run level 3: Full multi-user mode (text-based)
- Run level 4: Undefined or not used (reserved for user-defined settings)
- Run level 5: Full multi-user mode (with graphical user interface)
- Run level 6: Reboot the system



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